

CS 540 Spring 2025 Midterm

Exam Version: 1

First, Last Name:

Section:

Net ID: _____@wisc.edu

Campus ID Number:

1. You cannot sit next (front, back, left (within two seats including an empty seat), right (within two seats including an empty seat)) to someone you know. Switch seats now if that's the case.
2. Fill in these fields (left to right) on the scantron sheet using a pencil:
 - LAST NAME (family name) and FIRST NAME (given name), fill in bubbles
 - IDENTIFICATION NUMBER is your Campus ID number, fill in bubbles
 - Under A, B, C of SPECIAL CODES, fill in your three-digit section number (001, 002, 003)
 - **Under F of SPECIAL CODES, fill in Exam version above.** This is very important!

Make sure you fill all the above accurately in order to get graded.

3. Mark your answers on the Scantron sheet **and this handout**. Use a #2 pencil to mark all answers.
4. You may only reference your one-sheet notes. Please turn off and put away portable electronics now.
5. If you need to use the restroom, your phone must remain in the room, placed face down on your desk and kept visible to the proctors.
6. You have 90 minutes to take the exam. Once you have finished, please show your student ID to one of the proctors and submit both your scantron and this handout.

Good luck!

	A	B
C	0.3	0.1
D	0.4	0.2

Table 1: Joint Probability Distribution

- Given the above joint probability distribution table, what is $\mathbb{P}[B|C]$?
 - 1/10
 - 1/4
 - 3/10
 - 1/3
 - 2/5

Solution: $\mathbb{P}[B|C] = \mathbb{P}[B, C] / \mathbb{P}[C] = 0.1 / (0.3 + 0.1) = 1/4$

The correct choice is B

- Alice knows that for any email she receives, the probability it is spam is 0.4, and the probability it has an all-capitalized subject line is 0.8. She also knows that if an email is spam, the probability that the subject line is all capitalized is 0.6. If she sees an email in her inbox with an all-capitalized subject line, what is the probability that it is spam?
 - 6/25
 - 3/10
 - 2/5
 - 3/5
 - 16/25

Solution: Using Bayes' Rule,

$$\mathbb{P}[\text{spam}|\text{capitalized}] = \frac{\mathbb{P}[\text{capitalized}|\text{spam}]\mathbb{P}[\text{spam}]}{\mathbb{P}[\text{capitalized}]} = \frac{0.6 \cdot 0.4}{0.8} = \frac{3}{10}$$

The correct choice is B.

- Which one of the following is often used for classification?
 - Principal component analysis
 - Linear regression
 - Logistic regression
 - k-means clustering
 - None of the above

Solution: C. Logistic regression is used for binary classification. Options A and C are unsupervised learning algorithms.

4. What is true about PCA?
- A. It cannot be used for visualization.
 - B. We can project data on m axis that can be found using the bottom- m eigenvectors of the covariance matrix.
 - C. Data needs to have labels.
 - D. The goal is to minimize projection error.
 - E. None of the above.

Solution: D is clearly stated on slide 19 in lecture 4.

5. In what application would you use an unsupervised learning algorithm instead of a supervised learning algorithm?
- A. Predicting whether an image contains a dog or not.
 - B. Deciding which courses have similar content.
 - C. Binary classification.
 - D. Predicting what the weather will be tomorrow.
 - E. None of the above

Solution: B. Any prediction/classification-based task would be supervised learning. On the other hand, finding a pattern or grouping similar items would be unsupervised.

6. Which of the following activation function always produces output value in $[0, 1]$?
- A. ReLU
 - B. tanh
 - C. sigmoid
 - D. linear
 - E. None of the above

Solution: C. ReLU and linear can be arbitrarily large. Tanh function normalizes output between -1 and 1. Sigmoid normalizes outputs between 0 and 1.

7. You are performing PCA on a dataset of points in $\mathbb{R}^3$. One of the original data points is $x = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$. Given that the **second** principal component is $v_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ and the **third** principal component is $v_3 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$, what is the compressed representation of x after projection onto the **first two** principal components?
- A. $\begin{bmatrix} \frac{2}{\sqrt{2}} \\ 0 \end{bmatrix}$
- B. $\begin{bmatrix} 0 \\ \frac{2}{\sqrt{2}} \end{bmatrix}$
- C. $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$
- D. $\begin{bmatrix} 2 \\ 0 \end{bmatrix}$
- E. None of the above.

Solution: Since the principal components must be orthogonal to each other, we can deduce that the first principal component is $v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$. Then the compressed representation of x is $\begin{bmatrix} v_1^\top x \\ v_2^\top x \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$. The correct answer is **D**.

8. Consider the matrix $A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}$. A has an eigenvector $v = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. What is the corresponding eigenvalue λ ?
- A. $\lambda=5$
- B. $\lambda=3$
- C. $\lambda=6$
- D. $\lambda=7$
- E. None of the above

Solution: D. We compute

$$\begin{aligned}Av &= \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 7 \\ -14 \end{bmatrix} \\ &= 7 \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ &= 7v,\end{aligned}$$

so $\lambda = 7$.

9. We have three boolean variables: P , Q , and R . Suppose P is false, Q is True, and R is True. Which of the following is **NOT** true?

- A. $\neg(P \vee Q) \iff (\neg P \wedge \neg Q)$
- B. $(P \vee \neg Q) \implies ((R \wedge Q) \implies (\neg Q))$
- C. $\neg(R \vee Q) \iff (Q \wedge R)$
- D. $(P \vee (Q \wedge R)) \implies Q$
- E. None of the above

Solution: C is not true. Choice A evaluates to true. This is de Morgan's law, but we can also see that $\neg(P \vee Q)$ is false and that $\neg P \wedge \neg Q$ is false, so the $\iff$ is true. Choice B evaluates to true. The left hand side is not true (P is not true and $\neg Q$ is not true) so the entire statement is true. Choice C is false. $\neg(R \vee Q)$ is false, so the $\implies$ direction is true. But $Q \wedge R$ is true so the $\impliedby$ direction is false, making the $\iff$ false. Choice D is true. Q and R are both true, so $Q \vee R$ is true. P is false. Combining the two, $P \wedge (Q \vee R)$ is true. And Q is true, so the $\implies$ is also true.

10. Which of the following best captures the purpose of word embeddings in NLP?

- A. To represent words as sparse, one-hot encoded vectors.
- B. To provide a dense vector representation that captures semantic similarity.
- C. To count the frequency of words in a document.
- D. To model word sequences using a bigram approach.
- E. To convert raw text into binary feature representations.

Solution: The correct answer is **(B)**.

11. You are building a language model using the n-gram approach. Which of the following statements is true?
- A. Smoothing is generally more useful for a 3-gram model than a 6-gram model.
 - B. A trigram model captures longer-range dependencies than a bigram model.
 - C. In a unigram model, the probability of a word depends on the previous word.
 - D. In an n-gram model, increasing n always makes probability estimation easier.
 - E. None of the above is true.

Solution: B. A trigram (3-gram) considers two previous words, whereas a bigram (2-gram) considers only one. This allows the trigram model to capture longer-range dependencies.

12. Given input $\mathbf{x}$, a perceptron outputs $\sigma(\mathbf{w}^\top \mathbf{x} + b)$. Let $\mathbf{w} = (1, 1)^\top$, $b = -1.5$, and σ is ReLU. What is the perceptron's output for input $\mathbf{x} = (0, 0)^\top$, $\mathbf{x} = (1, 0)^\top$, and $\mathbf{x} = (1, 1)^\top$, respectively?
- A. 0, 0, 0.5
 - B. 0, 0, 1
 - C. 0, 1, 2
 - D. -1.5, -0.5, 0.5
 - E. None of the above

Solution: A. Check the first input: $\mathbf{w}^\top \mathbf{x} + b = 0 - 1.5 = -1.5$. Relu is 0 for negative values so the first input gives 0. Then check the rest: $\mathbf{w}^\top \mathbf{x} + b = 1 - 1.5 = -0.5$ and $\mathbf{w}^\top \mathbf{x} + b = 2 - 1.5 = 0.5$. So the second output is also 0 and 0.5 for the third.

13. Let the corpus be (assume the symbols are read from left to right)

✱ ✂ ♥ ✂ ✱ ↗ ♥ ✱ ✱ ↗

Using a frequency estimate (i.e. maximum likelihood estimate), estimate the bigram probability $P(\uparrow \mid \star)$.

- A. 0
- B. 1/4
- C. 1/3
- D. 1/2
- E. None of the above

Solution: D. Count up the number of times that * appeared: 4. Count up the number of times that * appeared immediately after * appeared: 2. The MLE estimate is $2/4 = 1/2$.

14. We want to perform hierarchical clustering with two clusters for the following points: 0, 1, 10, 18, 20, 25. What are the clusters if we use single-linkage, complete-linkage, and average-linkage?
- A. Single linkage: (0, 1, 10), (18, 20, 25)
 Complete linkage: (0, 1), (10, 18, 20, 25)
 Average linkage: (0, 1, 10), (18, 20, 25)
 - B. Single linkage: (0, 1, 10, 18, 20), (25)
 Complete linkage: (0), (1, 10, 18, 20, 25)
 Average linkage: (0), (1, 10, 18, 20, 25)
 - C. Single linkage: (0, 1), (10, 18, 20, 25)
 Complete linkage: (0, 1), (10, 18, 20, 25)
 Average linkage: (0, 1), (10, 18, 20, 25)
 - D. Single linkage: (0, 1), (10, 18, 20, 25)
 Complete linkage: (0, 1), (10, 18, 20, 25)
 Average linkage: (0, 1, 10), (18, 20, 25)
 - E. None of the above

Solution: E. We can find the two clusters by iterating through clustering with each linkage method:

Single Linkage:

Iteration 1: (0, 1), 10, 18, 20, 25
 Iteration 2: (0, 1), 10, (18, 20), 25
 Iteration 3: (0, 1), 10, (18, 20, 25)
 Iteration 4: (0, 1), (10, 18, 20, 25)

Complete Linkage:

Iteration 1: (0, 1), 10, 18, 20, 25
 Iteration 2: (0, 1), 10, (18, 20), 25
 Iteration 3: (0, 1), 10, (18, 20, 25)
 Iteration 4: (0, 1, 10), (18, 20, 25)

Average Linkage:

Iteration 1: (0, 1), 10, 18, 20, 25
 Iteration 2: (0, 1), 10, (18, 20), 25
 Iteration 3: (0, 1), 10, (18, 20, 25)
 Iteration 4: (0, 1, 10), (18, 20, 25)

15. What is the k-means centroid of the cluster $\{1, \sqrt{2}, 3, 4, -\sqrt{2}, 4\}$?

- A. 3
- B. 15
- C. $\sqrt{6}$
- D. 2
- E. None of the above.

Solution: D. The centroid for the cluster is the average of the points in the cluster:

$$\frac{1 + \sqrt{2} + 3 + 4 - \sqrt{2} + 4}{6} = \frac{12}{6} = 2$$

16. Which of the statements is true?

- A. The K-means algorithm always converges and finds the globally optimal solution given a dataset and a fixed number of cluster centers.
- B. The K-nearest neighbors algorithm can be used in a purely unsupervised fashion in an email spam classification problem.
- C. Hierarchical clustering does not require the user to provide the number of clusters as an input.
- D. Spectral clustering cannot be used in situations where the data is not linearly separable.
- E. All the above are true.

Solution: Solution. (C) Hierarchical clustering does not require the number of clusters as input.

- (A) Is not true because K-means need not find the globally optimal solution
- (B) K-nearest neighbors would need a dataset of spam labels, i.e. a supervised dataset for the problem of email spam classification
- (D) Spectral clustering can represent non linear decision boundaries
- (E) Some answers are false.

17. Which of the following statements is NOT TRUE?

- A. The output of a linear regression model can be given by $y = w^T x + b$ where $w, x \in \mathbb{R}^m$ and $b \in \mathbb{R}$
- B. The method of least squares attempts to minimize the loss $\sum_{i=1}^m (y_i - \hat{y}_i)^2$ where y is the target and $\hat{y}$ is the prediction of the linear regression model.
- C. Adding more parameters to a linear regression model always improves its generalization performance.

- D. The closed form solution for linear regression is $w = (X^T X)^{-1} X^T y$ where $X \in R^{m \times d}$ represents the data points and $y \in R^m$ represents the labels.
- E. Logistic regression is used to convert the real-valued outputs of a linear regression model into classification probabilities.

Solution: Answer (C). Adding more parameters to a network can worsen its generalization performance due to overfitting.

18. Increasing the value of K in the KNN algorithm generally has which of the following effects?
- A. Decreases sensitivity to noise.
 - B. Increases sensitivity to noise.
 - C. It has no effect on sensitivity to noise.
 - D. It makes the model completely ignore noise.
 - E. None of the above.

Solution: A larger K makes the classifier more robust to noise. Hence, the correct answer is the **(A)** choice.

19. What key assumption does the Naive Bayes classifier make when applied to text classification tasks?
- A. All words in a document are interdependent.
 - B. Word order is essential for predicting the document's class.
 - C. Features (words) are conditionally independent given the class label.
 - D. The probability of each word is the same across all documents.
 - E. Only the presence of a word, not its frequency, is considered.

Solution: The correct answer is **(C)**.

20. You're training a simple linear regression model for predicting student study time based on past data from four UW–Madison students. The loss function is mean squared error (MSE), and the current gradient for each data point is as follows (these are precomputed for simplicity).

Student	Gradient
1	2
2	−1
3	3
4	−2

Suppose your learning rate is 0.1. What is the parameter update if you use mini-batch gradient descent with a batch size of 2, using Students 1 and 2 in the first batch?

- A. Update = $0.1 \times \frac{2+(-1)}{2} = 0.05$
- B. Update = $0.1 \times \frac{2-(-1)}{2} = 0.15$
- C. Update = $0.1 \times \frac{2+(-1)}{4} = 0.0167$
- D. Update = $0.1 \times (2 + (-1)) = 0.1$

Solution: The correct answer is **(A)**. We are using **mini-batch gradient descent** with:

- **Batch size** = 2
- **Learning rate** = 0.1
- First mini-batch includes **Students 1 and 2**

From the table, their gradients are:

Student 1: 2, Student 2: -1

In mini-batch gradient descent, we compute the average gradient over the batch:

$$\text{Average Gradient} = \frac{2 + (-1)}{2} = \frac{1}{2} = 0.5$$

Using the gradient descent update rule:

$$\text{Update} = \text{Learning Rate} \times \text{Average Gradient}$$

$$\text{Update} = 0.1 \times 0.5 = 0.05$$

21. Which of the following is a drawback of the KNN algorithm?

- A. It cannot be used for regression.
- B. It makes strong assumptions about the underlying data distribution.
- C. It cannot be used for classification tasks.
- D. It is sensitive to the choice of K .
- E. None of the above.

Solution: The correct answer is the **(D)** choice. If K is small, it can be sensitive to noise; if it is too large, it can decrease flexibility.

22. A 4-layer MLP has 400 trainable parameters, including biases. The model takes an input of size 4. The hidden layers have sizes 15, 10, and 10, respectively. What is the size of the model output?
- A. 5
 - B. 9
 - C. 10
 - D. 11
 - E. None of the above.

Solution: Let x denote the size of the output. Then the number of trainable weights in the MLP is

$$(4 \cdot 15 + 15) + (15 \cdot 10 + 10) + (10 \cdot 10 + 10) + (10x + x) = 345 + 11x.$$

Setting this equal to 400 yields $x = 5$. The correct answer is **A**.

23. Which of the following are important criteria for an activation function in neural networks?
- A. Non-linearity
 - B. Element-wise operation
 - C. Differentiability (almost everywhere)
 - D. Computational efficiency
 - E. All of the above

Solution: E.

24. Consider a simple neural network with a single output neuron using the sigmoid activation function $\sigma(x) = \frac{1}{1+e^{-x}}$. The network is trained using binary cross-entropy loss:

$$L(y, \hat{y}) = -[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})]$$

where y is the true binary label (0 or 1) and $\hat{y}$ is the predicted probability.

If we need to compute the gradient of this loss function with respect to the neuron's pre-activation value z (where $\hat{y} = \sigma(z)$), what is the correct expression for $\frac{\partial L}{\partial z}$?

- A. $\hat{y} - y$
- B. $y - \hat{y}$
- C. $-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$
- D. $-y(1 - \hat{y}) + (1 - y)\hat{y}$
- E. $-y(1 - \hat{y}) - (1 - y)\hat{y}$

Solution: A or D. Starting with the binary cross-entropy loss:

$$L(y, \hat{y}) = -[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})]$$

We want to find $\frac{\partial L}{\partial z}$, where $\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$

Using the chain rule:

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z}$$

First, let's find $\frac{\partial L}{\partial \hat{y}}$:

$$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}} \times (-1) = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$$

Next, we need $\frac{\partial \hat{y}}{\partial z}$, which is the derivative of the sigmoid function:

$$\frac{\partial \hat{y}}{\partial z} = \frac{\partial \sigma(z)}{\partial z} = \sigma(z) \times (1 - \sigma(z)) = \hat{y}(1 - \hat{y})$$

Now, we multiply these together:

$$\frac{\partial L}{\partial z} = \left[-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right] \times [\hat{y}(1 - \hat{y})] \tag{1}$$

$$= -y(1 - \hat{y}) + (1 - y)\hat{y} \tag{2}$$

$$= -y + y\hat{y} + \hat{y} - y\hat{y} \tag{3}$$

$$= -y + \hat{y} \tag{4}$$

$$= \hat{y} - y \tag{5}$$

Therefore, the gradient of the binary cross-entropy loss with respect to z is simply $\hat{y} - y$.

If you chose A or D you have received full credit.

25. In general, underfit models have

- A. Low training loss and high generalization loss.

- B. High training loss and high generalization loss.
- C. Low training loss and low generalization loss.
- D. High training loss and low generalization loss.

Solution: B, we should expect an underfit model to have high training and generalization loss.